Analyzing E-Learning Platform Purchases

Summary Report & Query Answers | Data Analytics — Module-End 3

Overview

The dataset covers 5 learners, 5 courses across three categories (Beginner, Intermediate, Advanced), and 8 purchase transactions recorded between May and August 2026. Total platform revenue across this sample is ₹1,12,000, at an average of ₹22,400 per learner.

Key Insights

- **Beginner courses dominate revenue:** The Beginner category drives most of the business — ₹78,000 in revenue (about 70% of the total) from 3 unique learners, compared to ₹24,000 from Intermediate (Python) and ₹10,000 from Advanced (SQL).
- **Machine Learning is the top-performing course:** The Machine Learning course (₹12,000/unit, Beginner category) is the single strongest performer, leading both in units sold (4) and revenue generated (₹48,000), despite being the most expensive course on the platform.
- **Spending is concentrated in a few learners:** Arun, Christ, and Deepak (₹34,000 / ₹24,000 / ₹29,000 respectively) sit above the ₹22,400 average and classify as "High Value," while Vihaan and Bala fall into the "Medium Value" band — a 60/40 split.
- **No cross-category purchasing yet:** Every learner who bought more than one course stayed within the Beginner category. No learner has yet purchased across categories, pointing to an untapped cross-sell opportunity into Intermediate/Advanced courses.
- **Full catalog engagement:** All 5 courses in the catalog have at least one purchase — there is no unsold or dead inventory in the current dataset.

Recommendations

- **1. Drive cross-selling:** Target existing Beginner-tier learners (especially repeat buyers like Vihaan, Arun, and Deepak) with bundled offers or discounts on SQL and Python to convert them into cross-category customers.
- **2. Double down on the star course:** Use Machine Learning's strong pull as a flagship offering — feature it in promotions and consider tiered pricing to sustain volume at its premium price point.
- **3. Nurture Medium Value learners:** Design a loyalty or re-engagement campaign aimed at moving Medium Value learners such as Vihaan and Bala toward higher spending tiers.
- **4. Improve data quality controls:** Standardize category labels at the data-entry stage (e.g., "Beginner" vs. "beginner" appeared inconsistently in the source data) to keep future category-level reporting accurate.

Query Answers

Data Exploration — Join Results

INNER, LEFT, and RIGHT JOIN versions all return the same 8 rows below, because every purchase links to a valid learner and course and every learner/course has at least one purchase — there are no orphan or unmatched records in this sample.

Learner	Course	Category	Qty	Total Amount (₹)	Purchase Date
Arun	Machine Learning	beginner	2	24,000.00	2026-05-01
Christ	Phython	intermediate	3	24,000.00	2026-05-12
Deepak	Machine Learning	beginner	2	24,000.00	2026-08-15
Vihaan Singh	Excel for beginners	Beginner	2	10,000.00	2026-07-01
Bala	SQL	Advanced	1	10,000.00	2026-05-03
Arun	Powerbi	beginner	2	10,000.00	2026-08-18
Vihaan Singh	Powerbi	beginner	1	5,000.00	2026-06-01
Deepak	Excel for beginners	Beginner	1	5,000.00	2026-05-10

Q1. Total Spending per Learner (with Country)

Learner	Country	Total Spending (₹)
Arun	India	34,000
Deepak	India	29,000
Christ	India	24,000
Vihaan Singh	India	15,000
Bala	India	10,000

Q2. Top 3 Most Purchased Courses (by Quantity)

Excel, Power BI, and Python are tied at 3 units each behind Machine Learning; add a secondary ORDER BY course_id (or course_name) for a deterministic tie-break in production.

Rank	Course	Total Quantity Sold
1	Machine Learning	4
2	Excel for beginners	3
3	Powerbi	3
—	Phython (tied at 3, not shown by LIMIT 3)	3

Q3. Category Revenue & Unique Learners

Category	Total Revenue (₹)	Unique Learners
Beginner	78,000	3
Intermediate	24,000	1
Advanced	10,000	1

Q4. Learners Who Purchased From More Than One Category

Result: empty set. Every learner with multiple purchases (Vihaan, Arun, Deepak) bought within the Beginner category only — no one has crossed into Intermediate or Advanced yet.

Q5. Courses Never Purchased

Result: empty set. All 5 courses in the catalog have at least one recorded purchase.

Q6. Learners Above Average Total Spending

Average learner spending = ₹22,400.

Learner	Total Spending (₹)
Arun	34,000
Deepak	29,000

Learner	Total Spending (₹)
Christ	24,000

Q7. Courses Priced Higher Than ANY Beginner-Category Course

Beginner-category prices are ₹5,000, ₹5,000, and ₹12,000, so ANY() checks against the minimum (₹5,000). Courses priced above ₹5,000 qualify — including Machine Learning itself.

Course	Unit Price (₹)
Machine Learning	12,000.00
SQL	10,000.00
Phython	8,000.00

Q8. Learners Above Their Country's Average Spending

All 5 learners are based in India, so with a single country in the sample this query returns the same 3 learners as Q6. The logic will differentiate correctly once multi-country data is added.

Learner	Total Spending (₹)
Arun	34,000
Deepak	29,000
Christ	24,000

Q9. CTE — Learners With Spending Above ₹10,000

Bala (exactly ₹10,000) is excluded because the filter is a strict greater-than.

Learner	Total Spending (₹)
Arun	34,000
Deepak	29,000
Christ	24,000
Vihaan Singh	15,000

Q10. CASE — Spending Classification

Learner	Total Spending (₹)	Spending Category
Arun	34,000	High Value
Deepak	29,000	High Value
Christ	24,000	High Value
Vihaan Singh	15,000	Medium Value
Bala	10,000	Medium Value

Q11. NULL Handling — Purchase Count per Course

COALESCE(COUNT(...), 0) guarantees a 0 instead of NULL for any course with no purchases; in this dataset every course already has at least one, so no zeros appear.

Course	Purchase Count
Excel for beginners	2
Powerbi	2
SQL	1
Phython	1
Machine Learning	2

Q12. View — category_performance_view

Category	Total Revenue (₹)	No. of Purchases	Avg Revenue/Purchase (₹)
Beginner	78,000	6	13,000.00
Intermediate	24,000	1	24,000.00

Category	Total Revenue (₹)	No. of Purchases	Avg Revenue/Purchase (₹)
Advanced	10,000	1	10,000.00